

River plastic outflows and LSAP communities

This report explains the source data, executed workflow and interpretation of river plastic outflows and lsap communities. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (Meijer et al., 2021, data set and article).

Map P06 | Documentation date 6 October 2026 | Units Metric tonnes/year | Model published 2021

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Modelled annual macroplastic outflow · midpoint scenario · tonnes/year



Circle area is proportional to emission. Teal dots are LSAP members. Distances use the supplied member coordinates, whose precision is unverified.

Meijer et al. 2021 · doi:10.6084/m9.figshare.14515590 · CC BY 4.0

LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P06

Figure P06 World overview exported from the completed ArcGIS Pro layout.

Circle area is proportional to emission. Teal dots are LSAP members. Distances use the supplied member coordinates, whose precision is unverified.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (Meijer et al., 2021, data set and article).

1. Download and preserve the Meijer et al. GIS shapefile and metadata; copy 31,819 outfalls into the analysis geodatabase.
2. Copy dots_exten into Emission_t_yr: midpoint modelled annual floating macroplastic emission in metric tonnes/year.
3. Generate geodesic near tables between copied LSAP member points and outfalls. Summarize distinct member counts within 20 km and 100 km for each outfall; join counts to the source copy.
4. Use proportional circles with area proportional to Emission_t_yr; use the pollution palette and translucent fills with dark edges. Overlay teal LSAP dots and ecclesiastical outlines.
5. Create world and Southeast Asia layouts with popups showing exact emissions and member counts.

Output fields and files

Field or band	Meaning
Emission_t_yr	Midpoint annual macroplastic emission; copied from dots_exten
Members20 and Members100	Distinct LSAP members within 20 and 100 km

Analysis dataset: PollutionAnalysis.gdb/River_Plastic_Outfalls. Styles/P06.lyrx and Styles/P06.mapx preserve layer and map definitions. Map_Exports/P06_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

An additional Southeast Asia detail layout and P06_Southeast_Asia_detail.png show the relation between proportional emission circles and member dots. Circle diameter is $40 \times \sqrt{\text{emission} / \text{maximum emission}}$ points; circle area is proportional to emission.

Interpretation and data handling

The model estimates emissions to the ocean at outfalls. Geographic proximity is not watershed connectivity, causal contribution or exposure. Meijer GIS dataset, CC BY 4.0.

The Meijer midpoint GIS data are CC BY 4.0. Retain the authors, data DOI, scenario and units with any reused source or derived map. The original shapefile and copied outfall layer remain in the project.

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

References

Meijer, L., van Emmerik, T., van der Ent, R., Schmidt, C., & Lebreton, L. (2021). Supplementary data for More than 1000 rivers account for 80 percent of global riverine plastic emissions into the ocean (Version 1) [Data set]. Figshare. Midpoint emissions shapefile. [Source link](#)

Meijer, L. J. J., van Emmerik, T., van der Ent, R., Schmidt, C., & Lebreton, L. (2021). More than 1000 rivers account for 80 percent of global riverine plastic emissions into the ocean. Science Advances, 7(18), eaaz5803. [Source link](#)

Esri. (n.d.). Generate Near Table. ArcGIS Pro documentation. Reference for the geodesic proximity tools used in this project. Accessed 6 October 2026. [Source link](#)

LSAP Pollution Atlas. (2026). Internal source copies, executed analysis scripts and validation records [Project records]. LSAP_Pollution.aprx; FoundationCopies.gdb; PollutionAnalysis.gdb; foundation_manifest.json; raster_audit.json; freshwater_aggregate_QA.json; waste_QA.json; cleanup_audit.json; project_QA.json. Project root: Atlas Analyses/Pollution. Internal provenance is distinct from published source citations.